

which is much more flexible than was possible before. RTM is possible with the newest XBEGIN, XEND and XABORT commands and also allows developers to specify instructions that can be executed upon the failure of a particular transaction. The abort information is returned in the EAX register where it can be accessed easily. The XTEST instruction also allows you to check if a transactional region is being executed or not.

OTHER EXTENSIONS

Intel has added a third set of instructions called Bit Manipulation Instructions (BMI), which include 15 new instructions for handling scalar bit fields and operates on general purpose registers. These mainly have an application in Cryptography, Hashing and large-number Arithmetic. Haswell also adds a new Big Endian Move Instruction called MOVBE which is used to provide bi-endian support and allows for swapping of high and low bits of a long value.

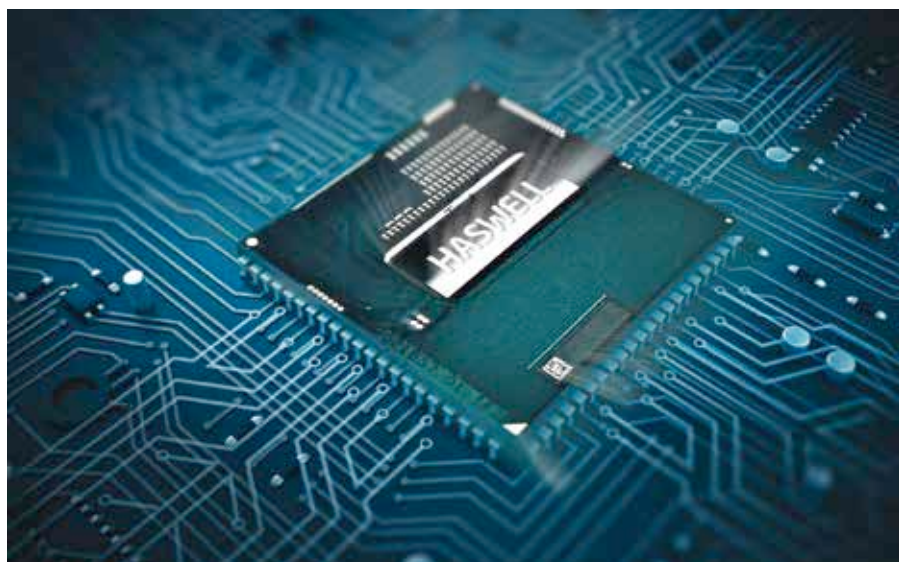
INTEL VT (VIRTUALIZATION) IMPROVEMENTS

Haswell has also significantly improved virtualization performance including guest-host transition times and a new VMFUNC instruction which allows for hyper-calls without a vmexit. There is also a new accessed and dirty bit for extended page tables to avoid vmexits.

GRAPHICS

The Intel on-board graphics solution was moved along with the processor chip, so that it can be bundled as processor graphics with much more powerful graphics capabilities. Haswell's GPU will be available in three configurations levels i.e. GT1, GT2 and GT3. There is another level called GT3e which will be available with embedded DRAM. This will offer large performance gains in operations where memory bandwidth has always been a problem. According to Intel, such operations include particle blending, post-processing, and other "write-then-read" operations, like shadow or reflection map rendering.

Some of these top-end features will only be offered with the Intel Iris Pro graphics solution, which is expected to be included in some of the high performing expensive chipsets. These chips are touted as being able to double the graphics performance as compared to previous generations of the GPU, thus providing serious competition for third-party graphics



Haswell chips have made several improvements over the previous gen

chipsets of Nvidia and ATI. There are basically graphics solutions being offered in five tiers which include Intel HD, Intel HD Graphics 4600, Intel HD Graphics 5000, Intel Iris 5100 and Iris Pro 5200. Each of these can be mapped to some or the other configuration level of the GPU. Intel Graphics now also offers support for DirectX 11.1, OpenCL 1.2, OpenGL 4.1, screen collage display and enhanced 4Kx2K playback.

VIDEO CODEC IMPROVEMENTS

Haswell includes support for hardware level Scalable Video Codec which allows lower resolution bitstreams to be derived from higher quality versions. This is expected to be beneficial for multi-party video conferencing applications. The motion JPEG hardware decoder and MPEG2 encoder allows for further benefits in video applications such as HD video conferencing and DLNA streaming. The Video processing functions have also been given an overhaul with a newly designed Video Quality Engine and availability of newer video processing functions such as Gamut Expansion, Skin Tone Tuned Image Enhancement Filter, Frame Rate Conversion and Image Stabilization. The newer Video Quality Engine almost doubles the throughput for video processing functions.

INTEL WIDI

The Intel WiDi or Wireless Display technology allows for wireless streaming of movies, games, presentations or other content from an ultrabook to a WiDi-enabled HDTV screen for

a better viewing experience. WiDi only requires the WiDi driver software on the computer and a dongle or adaptor for the HDTV if it doesn't have an integrated WiDi solution. Once all these requirements are met, WiDi can be connected to stream the whole operating system screen including the desktop in resolution ranging from 720p to 1080p. If any audio output is available, it is also streamed in DVD quality that supports upto 5.1 channel surround sound.

But you might wonder that if such technologies for connectivity with a second screen have existed for a long time, then what purpose does WiDi solve? Until now, most of the existing solutions have been limited to wired connections like HDMI or to a defined set of proprietary services which are unable to provide much freedom outside that specific application. WiDi provides a full wireless streaming solution for the complete desktop and is made to cater to new age devices such as ultrabooks etc. It even allows for never before seen features in wireless streaming such as extended display, which can for example allow you to have two screens – one for the streaming device and one for the remote display. This means one can possibly watch a movie being streamed to the television, while at the same time, work on a Word document! It works with the help of Intel's My WiFi technology, which appears to be some form of Wi-Fi direct connectivity between two devices. This connection used to broadcast content rendered through the Intel Centrino chipsets over the existing Wi-Fi connection, so that a second display or a mirrored display is available to the user.